

Evaluation Framework and Updated Results for LLM Dental Patient Simulation

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1 Evaluation Methodology

To ensure the clinical accuracy and immersive quality of the virtual patient, we implemented a rigorous testing suite comprising three distinct metrics. The model was subjected to a standardized interview protocol consisting of opening questions, chief complaint inquiries, and disease-specific diagnostic questions.

1.1 Symptom Fidelity Score (SFS)

The Symptom Fidelity Score measures the accuracy of the LLM’s clinical presentation against the ground truth symptoms defined in the knowledge base. It penalizes both the omission of required symptoms and the hallucination of non-existent symptoms.

Let $S_{expected}$ be the set of symptoms required for a specific diagnosis. Let $S_{mentioned}$ be the set of symptoms extracted from the LLM’s responses, and $S_{hallucinated}$ be the subset of mentioned symptoms that are medically contraindicated or irrelevant.

$$SFS = \max \left(0, 1 - \frac{N_{missing} + N_{hallucinated}}{|S_{expected}|} \right) \quad (1)$$

1.2 Role-Playing Consistency Score (RPCS)

This metric evaluates the stability of the patient persona. It detects breaks in character, such as the use of medical jargon, providing medical advice, or meta-awareness (“I am an AI model”).

$$RPCS = \begin{cases} 1.0 & \text{if } V = 0 \\ 0.8 & \text{if } 0 < R_v \leq 0.1 \\ 0.5 & \text{if } 0.1 < R_v \leq 0.3 \\ 0.2 & \text{if } R_v > 0.3 \end{cases} \quad (2)$$

1.3 Clinical Realism Score (CRS)

The CRS evaluates linguistic and behavioral realism across four dimensions: subjective symptom description, proper layperson vocabulary, pain description specificity, and internal consistency.

$$CRS = \frac{1}{16} \sum_{i=1}^4 S_{dim_i} \quad (3)$$

2 Model Specification

All simulations were generated using the large language model **meta-llama/llama-4-scout-17b-16e-instruct**, executed via the Groq API.

The model operates with:

- 17B parameters
- 16E token representation

- high-throughput, low-latency inference through Groq hardware
- no fine-tuning (pure prompt-based conditioning)
- custom system prompts enforcing patient persona, symptom constraints and dialogue consistency

The evaluation framework interacts with the model in a controlled step-by-step simulation environment, where each LLM reply undergoes automatic parsing for symptom extraction, hallucination detection and persona consistency scoring.

3 Experimental Results

The updated model was evaluated across 8 dental pathologies. For each condition, 2 full consultation simulations were run (Total $N = 16$).

3.1 Aggregate Performance

The model demonstrated robust performance across all metrics, with particularly strong results in Clinical Realism and Role Consistency.

Table 1: Aggregate Metrics Across All Tests

Metric	Mean	Min	Max
Symptom Fidelity (SFS)	0.984	0.875	1.000
Role Consistency (RPCS)	0.950	0.800	1.000
Clinical Realism (CRS)	0.981	0.938	1.000

3.2 Performance by Disease

Table 2: Performance Breakdown by Disease (Average of 2 runs)

Disease	Avg SFS	Avg RPCS	Avg CRS	Overall Score
Chronic Apical Periodontitis	1.000	1.000	1.000	1.000
Pericoronitis	1.000	1.000	1.000	1.000
Reversible Pulpitis	1.000	1.000	1.000	1.000
Irreversible Pulpitis	1.000	1.000	1.000	1.000
Acute Apical Periodontitis	1.000	1.000	0.938	0.979
Pulp Necrosis	1.000	0.900	0.969	0.969
Simple Cavities	1.000	0.800	1.000	0.933
Periodontal Abscess	0.875	0.900	0.938	0.904

3.3 Error Analysis

- **Symptom Fidelity:** Minor reductions in SFS appeared primarily in *Periodontal Abscess*, where the mapping contains a large number of required symptoms, many of which real patients do not spontaneously mention.
- **Hallucinations:** No significant hallucinations were observed across most diseases. The model showed strong discipline in avoiding symptoms unrelated to the diagnosis.
- **Role Consistency Issues:** Occasional RPCS reductions (to 0.8) were caused by very small meta-awareness artifacts such as conversational fillers that match predefined violation patterns.
- **Clinical Realism:** CRS remained consistently high across all pathologies, with scores between 0.938 and 1.000, indicating strong alignment with natural patient speech patterns.